TECHNICAL DILIGENCE REVIEW - RECORD 014

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-014

1. Scope of the delivery-process review and its limits

Scope of the present report is the engineering delivery process, pipeline metrics and change-management controls, and the cut-off for evidence is 2026-03-03. As instructed, Pinehurst Systems Assurance confined TDR-2026-05 to the engineering delivery process, pipeline metrics and change-management controls; three adjacent questions raised during the work are listed as follow-ups rather than answered here. Work on this report ran over 7 weeks and included 12 interviews and read-only access to the engineering repositories. Method in this report was three-fold: read the design record, interview the owning engineers, and verify the claim against a running system. Cross-references in this report point to the evidence index rather than to a summary of the evidence.

Access for the report at TDR-2026-05 was sufficient for the conclusions drawn; where it was not, the limitation is stated at the point it bites. K. Vandermeer was granted read access to the code, the pipeline and the incident tracker for the purposes of report TDR-2026-05; production access was not requested. Ardent Peak commissioned TDR-2026-05 from Pinehurst Systems Assurance to test the engineering delivery process, pipeline metrics and change-management controls ahead of the Series B financing. The subject of the present report is the engineering delivery process, pipeline metrics and change-management controls, examined for Ardent Peak by Pinehurst Systems Assurance over 7 weeks ending 2026-03-03. K. Vandermeer notes that the company answered every document request, though two answers arrived after the draft.

TDR-2026-05 carries one structural limitation: the company's own metrics are the only source for several objectives, and the reviewer could not build an independent measurement inside the engagement. The principal limitation on this report is time: 7 weeks was enough to test the control path and not enough to test the reporting path. Two positions in TDR-2026-05 are unresolved between the reviewer and the engineering team, and both are left unresolved on the page. Not every finding in report TDR-2026-05 is accepted by management: 2 are declined, and the declinatures are reproduced rather than argued away. The company reviewed a draft of TDR-2026-05 for factual accuracy and its corrections are reflected without altering any conclusion.

For TDR-2026-05 the reviewer sampled 7 sites, including the largest deployment and the newest. Selection for this report favoured components that appear in the incident record, which biases the sample towards known trouble and is stated as such. None of the findings in this report would in the reviewer's view justify withholding completion, and two would justify a specific undertaking. Pinehurst Systems Assurance notes that 3 of the matters in this report are capable of being closed before completion if the company is asked in time. K. Vandermeer has stated the limitations at the point they bite rather than collecting them into a disclaimer.

Where the present report says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. Report TDR-2026-05 grades evidence as strong, moderate or limited, and a limited grade is not upgraded because the consequence would be serious. Interview evidence in TDR-2026-05 came from 12 sessions across 9 teams, each minuted and circulated for correction. This report draws on 12 interviews, including the engineering owners of every component examined in detail. K. Vandermeer has not smoothed the sequence of this report; it follows the order in which the evidence arrived.

Pinehurst Systems Assurance asked the same four questions of every team interviewed for report TDR-2026-05, which is how the inconsistencies below were found. 9 teams contributed to the interview programme for the report at TDR-2026-05, and one of them was interviewed twice after a document arrived late. The overall conclusion of the present report is that the technical estate would support the business plan through the next twelve months and not through the fleet size the plan assumes for 2027 without the identified programme. On the evidence available by 2026-03-03, the present report concludes that the technical estate would support the business plan through the next twelve months and not through the fleet size the plan assumes for 2027 without the identified programme. Pinehurst Systems Assurance has reported the position as at 2026-03-03 and has not attempted to predict where it will be at completion.

Schedule U - Qualifications carried forward, with evidence strength (TDR-2026-05)

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
OQ-14-01	the supplier certificate was	limited	would change the	a document the company

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
	read rather than the test witnessed		conclusion if resolved adversely	has not produced
OQ-14-02	the supplier certificate was read rather than the test witnessed	strong	is reflected in the post-close plan	an independent measurement
OQ-14-03	two teams each believe the other owns this	moderate	does not change the conclusion	a decision from management
OQ-14-04	two teams each believe the other owns this	strong	would change the conclusion if resolved adversely	a re-run of the original test
OQ-14-05	the retention window closed before the raw data was requested	limited	would change the conclusion if resolved adversely	a document the company has not produced
OQ-14-06	the supplier certificate was read rather than the test witnessed	strong	leaves one section unconcluded	an independent measurement
OQ-14-07	two teams each believe the other owns this	limited	is reflected in the post-close plan	one further interview
OQ-14-08	the second bench rig does not cover this path	strong	leaves one section unconcluded	a document the company has not produced
OQ-14-09	the objective is met on the company's own measurement only	moderate	leaves one section unconcluded	an independent measurement
OQ-14-10	the population completeness could not be proven	strong	would change the conclusion if resolved adversely	a witnessed test
OQ-14-11	the compensating control was not tested	moderate	is reflected in the post-close plan	a witnessed test
OQ-14-12	the measurement basis changed during the reviewed period	limited	is reflected in the post-close plan	a witnessed test
OQ-14-13	the retention window closed before the raw data was requested	strong	would change the conclusion if resolved adversely	a re-run of the original test
OQ-14-14	the objective is met on the company's own measurement only	moderate	does not change the conclusion	an independent measurement
OQ-14-15	the population completeness could not be proven	strong	is reflected in the post-close plan	a document the company has not produced
OQ-14-16	two teams each believe the other owns this	moderate	narrows the conclusion to the tested scale	a re-run of the original test
OQ-14-17	the remediation landed without a regression test	limited	narrows the conclusion to the tested scale	one further interview
OQ-14-18	the remediation landed without a regression test	strong	narrows the conclusion to the tested scale	a decision from management
OQ-14-19	the objective is met on the company's own measurement only	moderate	narrows the conclusion to the tested scale	one further interview
OQ-14-20	the second bench rig does not cover this path	moderate	narrows the conclusion to the tested scale	a witnessed test
OQ-14-21	the second bench rig does not cover this path	limited	leaves one section unconcluded	an independent measurement
OQ-14-22	the population completeness could not be proven	moderate	would change the conclusion if resolved adversely	a decision from management
OQ-14-23	the retention window closed before the raw data was requested	limited	leaves one section unconcluded	a witnessed test
OQ-14-24	the measurement basis changed during the reviewed period	moderate	would change the conclusion if resolved adversely	a witnessed test
OQ-14-25	the remediation landed without a regression test	moderate	does not change the conclusion	a decision from management
OQ-14-26	the retention window closed before the raw data was requested	strong	leaves one section unconcluded	a witnessed test
OQ-14-27	the population completeness could not be	limited	is reflected in the post-close plan	an independent measurement

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
	proven			
OQ-14-28	the objective is met on the company's own measurement only	strong	leaves one section unconcluded	a decision from management
OQ-14-29	the measurement basis changed during the reviewed period	limited	narrows the conclusion to the tested scale	one further interview
OQ-14-30	the compensating control was not tested	strong	does not change the conclusion	a witnessed test
OQ-14-31	the compensating control was not tested	limited	leaves one section unconcluded	a decision from management
OQ-14-32	the supplier certificate was read rather than the test witnessed	strong	is reflected in the post-close plan	a re-run of the original test

2. Pipeline stages, durations and gate configuration

About build, robot firmware images the reviewer would add that the cross toolchain is rebuilt weekly, which produces a periodic slow build. Build, robot firmware images is responsible for more than the two preceding stages combined of the total pipeline time. Build, robot firmware images is Robot Firmware's to fix when it degrades. build, robot firmware images rests on a single source, and the reviewer has said so rather than implying corroboration.

In the merge path integration tests, cloud is blocking. The direction of travel on integration tests, cloud is that it improved in flakiness after two suites were quarantined. Ownership of integration tests, cloud sits with Fleet Services. integration tests, cloud is the entry that changed most between the first and second document requests behind TDR-2026-05.

Timing on contract tests, warehouse connectors is 9 m 40 s mean, 15 m 55 s at p90, measured over the reviewed quarter. The contract tests, warehouse connectors stage belongs to Integrations. Flakiness on contract tests, warehouse connectors stands at 3.7% against a total failure rate of 9.3%. contract tests, warehouse connectors has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Maintenance of simulation suite, short falls to Autonomy Core. Simulation suite, short carries this qualification: determinism problems in the physics core inflate the apparent failure rate. Simulation suite, short takes 33 m 10 s on average and 52 m 00 s at the 90th percentile. simulation suite, short is treated as material to the investment case rather than as a routine engineering item.

Simulation suite, nightly became slower and more reliable at the same time, and the team could explain why. For simulation suite, nightly the accountable team is Autonomy Core. Measured against lead time, simulation suite, nightly costs little on a clean run and a great deal on a contended one. simulation suite, nightly was tested rather than accepted, and the test is described in the working papers behind TDR-2026-05.

Maintenance of hardware-in-the-loop, bench falls to Robot Firmware. Hardware-in-the-loop, bench deserves one comment: queueing on two rigs is the main contributor to firmware lead time. Hardware-in-the-loop, bench was stable in every respect the reviewer could measure, and the team could explain why. hardware-in-the-loop, bench is the clearest instance in TDR-2026-05 of a control that exists on paper and is weaker in practice.

The dependency and licence scan stage belongs to Security Engineering. Dependency and licence scan improved in flakiness after two suites were quarantined, which the reviewer read from the pipeline's own history rather than from a summary. Measured against lead time, dependency and licence scan costs little on a clean run and a great deal on a contended one. dependency and licence scan was reviewed against the position twelve months earlier, and nothing measurable has changed.

Of the end-to-end lead time, secret scan accounts for more than the two preceding stages combined. Security Engineering owns secret scan, and the on-call for pipeline health is a separate rotation. Secret scan is blocking in the merge path. secret scan was raised again with the owning engineer after the initial interview, and the position did not change.

Schedule Q - Pipeline stage duration, failure rate and gate configuration (TDR-2026-05)

Pipeline stage	Mean duration	p90 duration	Failure rate	Flaky rate	Gate	Owning team
lint and format	1 m 10 s	1 m 40 s	3.1%	0.1%	blocking	Developer Productivity
unit tests, cloud services	6 m 20 s	9 m 05 s	5.4%	1.9%	blocking	Fleet Services
unit tests, robot software	11 m 45 s	18 m 30 s	7.8%	4.6%	blocking	Autonomy Core
static analysis, robot software	14 m 05 s	22 m 10 s	6.2%	0.4%	blocking	Autonomy Core
build, cloud container images	8 m 15 s	12 m 40 s	2.7%	0.3%	blocking	Platform Infrastructure
build, robot firmware images	19 m 30 s	31 m 20 s	4.1%	0.6%	blocking	Robot Firmware
integration tests, cloud	24 m 50 s	41 m 15 s	12.6%	8.2%	blocking	Fleet Services
contract tests, warehouse connectors	9 m 40 s	15 m 55 s	9.3%	3.7%	blocking	Integrations
simulation suite, short	33 m 10 s	52 m 00 s	8.9%	5.1%	blocking	Autonomy Core
simulation suite, nightly	3 h 40 m	4 h 25 m	14.2%	6.8%	advisory	Autonomy Core
hardware-in-the-loop, bench	1 h 15 m	2 h 05 m	10.4%	2.2%	blocking for firmware	Robot Firmware
dependency and licence scan	4 m 25 s	6 m 50 s	1.8%	0.2%	advisory	Security Engineering
secret scan	2 m 05 s	3 m 15 s	0.9%	0.1%	blocking	Security Engineering
container image vulnerability scan	5 m 35 s	8 m 20 s	16.7%	1.1%	advisory	Security Engineering
performance smoke, cloud	12 m 20 s	19 m 45 s	6.6%	3.4%	advisory	Platform Infrastructure
release packaging and signing	7 m 50 s	11 m 10 s	2.2%	0.2%	blocking	Developer Productivity

3. Failure and flake rates, and what they cost the team

Measured against lead time, container image vulnerability scan costs the largest single share of any stage. For container image vulnerability scan the accountable team is Security Engineering. The failure rate reflects new advisories rather than new defects is the finding attaching to container image vulnerability scan. container image vulnerability scan sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Advisory is the gate configuration for performance smoke, cloud. The failure rate on performance smoke, cloud is 6.6%, with 3.4% attributable to flakiness. Performance smoke, cloud contributes a share that is invisible because it runs in parallel to the change lead time. performance smoke, cloud is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Release packaging and signing carries this qualification: the signing step is the only stage requiring a human approval. Of the 2.2% of release packaging and signing runs that fail, 0.2% are reruns that pass unchanged. Ownership of release packaging and signing sits with Developer Productivity. release packaging and signing was raised again with the owning engineer after the initial interview, and the position did not change.

Gate status for lint and format is blocking. The failure rate on lint and format is 3.1%, with 0.1% attributable to flakiness. The material point on lint and format is that fast and stable; the only stage no engineer complains about. lint and format was excluded from an earlier draft of TDR-2026-05 and reinstated after the reviewer obtained the supporting record.

Fleet Services maintains unit tests, cloud services and answers for its reliability. The pipeline treats unit tests, cloud services as blocking. Unit tests, cloud services degraded after a new rule set was added, and no one the reviewer asked could account for it. unit tests, cloud services is the reason one section of TDR-2026-05 concludes narrowly rather than generally.

The direction of travel on unit tests, robot software is that it degraded after a new rule set was added. Ownership of unit tests, robot software sits with Autonomy Core. In lead-time terms unit tests, robot software contributes a small and

predictable share. unit tests, robot software is the entry that changed most between the first and second document requests behind TDR-2026-05.

The pipeline treats static analysis, robot software as blocking. Static analysis, robot software degraded after a new rule set was added across the measured window. In lead-time terms static analysis, robot software contributes a share that is invisible because it runs in parallel. static analysis, robot software would be a candidate for the first post-close engineering review under any reasonable plan.

Build, cloud container images is configured blocking, which is the fact that determines what can reach production. The reviewer's observation on build, cloud container images is that cached layers keep this stage predictable. The build, cloud container images stage runs to a mean of 8 m 15 s, with a 90th percentile of 12 m 40 s. build, cloud container images is included because a reader of TDR-2026-05 should be able to disagree with the inference without repeating the work.

Schedule R - Release history sampled across the reviewed period (TDR-2026-05)

Release	Date	Component	Gate set applied	Approver	Rollback	Outcome
R6.1-101	2025-07-30	Hardware-in-the-Loop Rig	full gate set	Grace Lindqvist	not required	promoted to ring one only
R6.0-102	2025-08-04	Battery Management Service	full gate set	Marta Kellerman	not required	withdrawn
R5.3-103	2025-08-09	Telemetry Ingest	gate set with the nightly suite waived	Theo Bassett	rolled back same day	promoted to ring one only
R8.13-104	2025-08-13	Safety Controller	full gate set	Yuki Harada	not required	paused at ring two
R7.1-105	2025-08-17	Telemetry Ingest	full gate set	Bettina Vogel	rolled back next day	promoted to ring one only
R4.7-106	2025-08-19	Fleet Allocator	gate set with the nightly suite waived	Sam Oyelaran	not required	promoted to all rings
R7.6-107	2025-08-25	Speed and Separation Monitor	full gate set	Yuki Harada	rolled back same day	promoted to all rings
R7.10-108	2025-08-28	Alarm Router	emergency path	Bettina Vogel	not required	promoted to all rings
R4.7-109	2025-09-01	Release Orchestrator	gate set with the nightly suite waived	Erik Palmgren	not required	promoted to all rings
R6.8-110	2025-09-04	Fleet Allocator	gate set with the nightly suite waived	Farida Cisse	not required	promoted to ring one only
R5.11-111	2025-09-09	Costmap Fusion	gate set with the nightly suite waived	Sam Oyelaran	not required	withdrawn
R7.9-112	2025-09-12	OTA Updater	full gate set	Colin Achebe	not required	promoted to all rings
R6.5-113	2025-09-17	Map Authoring Studio	emergency path	Nadia Semenova	not required	promoted to ring one only
R4.7-114	2025-09-21	Release Orchestrator	gate set with the bench suite waived	Daniel Okafor	not required	promoted to ring one only
R5.4-115	2025-09-26	Lidar Driver	full gate set	Ines Duarte	rolled back same day	paused at ring two
R6.10-116	2025-09-29	Site Provisioning Service	full gate set	Ines Duarte	not required	promoted to all rings
R7.7-117	2025-10-02	Identity and Authorization	full gate set	Marcus Duplessis	not required	withdrawn
R4.11-118	2025-10-08	Analytics Warehouse Loader	emergency path	Sam Oyelaran	not required	promoted to ring one only
R6.11-119	2025-10-12	Speed and Separation Monitor	gate set with the nightly suite waived	Nadia Semenova	rolled back same day	promoted to all rings
R4.7-120	2025-10-15	Edge Gateway Agent	gate set with the nightly suite waived	Colin Achebe	rolled back next day	withdrawn
R6.7-121	2025-10-19	Config Distribution	full gate set	Ines Duarte	not required	withdrawn

Release	Date	Component	Gate set applied	Approver	Rollback	Outcome
R6.7-122	2025-10-23	Order Broker	full gate set	Wanda Njoku	rolled back same day	paused at ring two
R8.3-123	2025-10-26	Release Orchestrator	gate set with the bench suite waived	Tomas Villareal	rolled back same day	promoted to all rings
R6.2-124	2025-10-30	Hardware-in-the-Loop Rig	full gate set	Farida Cisse	not required	withdrawn
R7.11-125	2025-11-05	Telemetry Ingest	gate set with the bench suite waived	Daniel Okafor	rolled back same day	promoted to ring one only
R7.5-126	2025-11-09	Costmap Fusion	emergency path	Wanda Njoku	not required	paused at ring two
R7.13-127	2025-11-11	Order Broker	full gate set	Ines Duarte	rolled back next day	withdrawn
R4.10-128	2025-11-17	Event Bus	gate set with the nightly suite waived	Grace Lindqvist	rolled back same day	withdrawn
R8.9-129	2025-11-21	OTA Updater	full gate set	Priya Raghunathan	not required	paused at ring two
R8.9-130	2025-11-24	WMS Connector	full gate set	Farida Cisse	not required	promoted to ring one only
R4.7-131	2025-11-27	OTA Updater	full gate set	Peter Halvorsen	not required	promoted to all rings
R6.9-132	2025-12-02	Simulation Harness	full gate set	Ines Duarte	not required	promoted to all rings
R4.5-133	2025-12-06	Speed and Separation Monitor	full gate set	Erik Palmgren	not required	promoted to all rings
R6.5-134	2025-12-10	Speed and Separation Monitor	gate set with the bench suite waived	Erik Palmgren	not required	withdrawn
R5.9-135	2025-12-13	Safety Controller	full gate set	Nathan Oduya	not required	promoted to all rings
R8.13-136	2025-12-19	Remote Operations Console	gate set with the nightly suite waived	Grace Lindqvist	not required	promoted to all rings
R5.7-137	2025-12-23	Motion Planner	gate set with the bench suite waived	Ines Duarte	not required	withdrawn
R7.2-138	2025-12-25	Billing Usage Meter	full gate set	Sam Oyelaran	not required	promoted to all rings
R4.8-139	2025-12-29	Order Broker	full gate set	Sonia Krall	not required	promoted to all rings
R7.0-140	2026-01-02	Site Provisioning Service	full gate set	Priya Raghunathan	rolled back same day	withdrawn
R6.1-141	2026-01-07	Traffic Manager	full gate set	Colin Achebe	not required	paused at ring two
R7.6-142	2026-01-11	Payload Scale Interface	gate set with the bench suite waived	Erik Palmgren	not required	promoted to all rings
R5.8-143	2026-01-14	Release Orchestrator	gate set with the nightly suite waived	Wanda Njoku	rolled back next day	paused at ring two
R5.12-144	2026-01-19	Config Distribution	gate set with the nightly suite waived	Colin Achebe	not required	promoted to all rings
R8.8-145	2026-01-24	Event Bus	full gate set	Anjali Sekhon	not required	withdrawn
R8.12-146	2026-01-27	Release Orchestrator	full gate set	Priya Raghunathan	not required	promoted to ring one only
R8.0-147	2026-01-31	Alarm Router	full gate set	Colin Achebe	not required	promoted to all rings
R4.8-148	2026-02-05	Gripper Controller	full gate set	Farida Cisse	not required	promoted to ring one only

3. Failure and flake rates, and what they cost the team - part 2 of 2

Set out next is the balance of the 3. Failure and flake rates, and what they cost the team, in the sequence in which the rows were originally compiled. The parts are to be read together.

Readers should carry the 3. Failure and flake rates, and what they cost the team's stated conventions across the break, since this part adds no definition of its own. The parts are to be read together.

Schedule R - Release history sampled across the reviewed period (TDR-2026-05) (continued, part 2 of 2)

Release	Date	Component	Gate set applied	Approver	Rollback	Outcome
R6.12-149	2026-02-08	Edge Gateway Agent	emergency path	Tomas Villareal	not required	paused at ring two
R7.8-150	2026-02-13	Gripper Controller	emergency path	Peter Halvorsen	not required	promoted to all rings
R6.3-151	2026-02-17	Traffic Manager	gate set with the bench suite waived	Farida Cisse	not required	promoted to all rings
R6.3-152	2026-02-21	Analytics Warehouse Loader	full gate set	Wanda Njoku	not required	paused at ring two
R7.12-153	2026-02-23	Video Snippet Service	full gate set	Erik Palmgren	not required	withdrawn
R6.10-154	2026-03-01	Telemetry Ingest	gate set with the nightly suite waived	Wanda Njoku	not required	promoted to all rings
R7.12-155	2026-03-03	Charge Scheduler	gate set with the bench suite waived	Grace Lindqvist	rolled back next day	paused at ring two
R5.10-156	2026-03-09	Site Provisioning Service	gate set with the nightly suite waived	Sonia Krall	not required	withdrawn
R7.5-157	2026-03-11	Telemetry Ingest	emergency path	Tomas Villareal	not required	promoted to all rings
R8.0-158	2026-03-17	Hardware-in-the-Loop Rig	gate set with the bench suite waived	Colin Achebe	not required	paused at ring two
R7.5-159	2026-03-21	Analytics Warehouse Loader	full gate set	Tomas Villareal	not required	promoted to ring one only
R5.4-160	2026-03-25	Edge Gateway Agent	gate set with the nightly suite waived	Colin Achebe	not required	withdrawn
R6.0-161	2026-03-27	Telemetry Ingest	full gate set	Nathan Oduya	not required	promoted to all rings
R6.7-162	2026-03-31	Event Bus	gate set with the nightly suite waived	Sam Oyelaran	rolled back next day	paused at ring two
R4.6-163	2026-04-06	Motion Planner	emergency path	Anjali Sekhon	not required	promoted to all rings
R7.3-164	2026-04-08	Release Orchestrator	full gate set	Farida Cisse	not required	promoted to all rings
R8.12-165	2026-04-13	Hardware-in-the-Loop Rig	full gate set	Ines Duarte	not required	promoted to ring one only
R7.12-166	2026-04-16	Order Broker	gate set with the nightly suite waived	Bettina Vogel	not required	promoted to all rings
R4.6-167	2026-04-21	Telemetry Ingest	gate set with the bench suite waived	Theo Bassett	not required	withdrawn
R5.11-168	2026-04-24	Billing Usage Meter	full gate set	Nadia Semenova	not required	promoted to all rings
R7.12-169	2026-04-30	Wheel Odometry Filter	gate set with the nightly suite waived	Erik Palmgren	not required	paused at ring two
R8.5-170	2026-05-03	Event Bus	gate set with the nightly suite waived	Marta Kellerman	not required	promoted to ring one only
R4.12-171	2026-05-08	Safety Controller	gate set with the nightly suite waived	Farida Cisse	rolled back same day	promoted to all rings
R5.8-172	2026-05-11	Payload Scale Interface	emergency path	Colin Achebe	rolled back next day	promoted to ring one only
R6.4-173	2026-05-16	Localization Engine	gate set with the nightly suite waived	Tomas Villareal	not required	promoted to all rings
R7.13-174	2026-05-19	Config Distribution	gate set with the nightly suite waived	Erik Palmgren	not required	withdrawn
R6.4-175	2026-05-24	Remote	full gate set	Ines Duarte	not required	paused at ring two

Release	Date	Component	Gate set applied	Approver	Rollback	Outcome
		Operations Console				
R4.13-176	2026-05-26	Config Distribution	gate set with the nightly suite waived	Theo Bassett	not required	withdrawn
R8.2-177	2026-05-30	Battery Management Service	full gate set	Owen Brantley	not required	paused at ring two
R6.13-178	2026-06-05	Fleet Allocator	gate set with the nightly suite waived	Erik Palmgren	not required	promoted to all rings
R7.0-179	2026-06-09	Wheel Odometry Filter	gate set with the nightly suite waived	Tomas Villareal	not required	paused at ring two
R8.0-180	2026-06-13	Traffic Manager	gate set with the bench suite waived	Marta Kellerman	not required	promoted to ring one only
R7.3-181	2026-06-16	Release Orchestrator	full gate set	Ines Duarte	not required	promoted to ring one only
R8.6-182	2026-06-20	Traffic Manager	full gate set	Theo Bassett	not required	promoted to all rings
R6.5-183	2026-06-25	Remote Operations Console	gate set with the nightly suite waived	Bettina Vogel	rolled back same day	withdrawn
R8.10-184	2026-06-28	Identity and Authorization	full gate set	Anjali Sekhon	not required	promoted to ring one only
R4.5-185	2026-07-02	OTA Updater	gate set with the bench suite waived	Bettina Vogel	not required	promoted to ring one only
R6.0-186	2026-07-07	Remote Operations Console	full gate set	Nathan Oduya	not required	withdrawn
R4.13-187	2026-07-10	Robot Shadow Store	full gate set	Bettina Vogel	rolled back same day	promoted to ring one only
R5.8-188	2026-07-15	OTA Updater	full gate set	Tomas Villareal	not required	promoted to all rings
R8.8-189	2026-07-18	Simulation Harness	full gate set	Bettina Vogel	rolled back next day	promoted to ring one only
R4.11-190	2026-07-22	Site Provisioning Service	gate set with the nightly suite waived	Sam Oyelaran	not required	paused at ring two
R8.10-191	2026-07-26	Alarm Router	gate set with the nightly suite waived	Marta Kellerman	rolled back same day	promoted to all rings
R5.3-192	2026-07-31	WMS Connector	gate set with the bench suite waived	Peter Halvorsen	not required	promoted to ring one only
R8.5-193	2026-08-02	Robot Shadow Store	gate set with the bench suite waived	Farida Cisse	not required	paused at ring two
R4.1-194	2026-08-06	Remote Operations Console	full gate set	Marcus Duplessis	rolled back same day	paused at ring two
R6.10-195	2026-08-11	Payload Scale Interface	gate set with the bench suite waived	Owen Brantley	not required	withdrawn
R8.12-196	2026-08-14	Localization Engine	full gate set	Wanda Njoku	not required	paused at ring two

4. Release cadence and ownership across the estate

Read against its objective, Alarm Router the design is sound and the test evidence is thinner than the design deserves. Against Alarm Router the backlog runs to 14 items, and it includes a known behaviour whereby an on-call rotation gap in the routing table falls back to a shared mailbox nobody watches at night. Implementation of Alarm Router rests on Python 3.11; the process runs on cloud, containers plus a scheduled dispatcher and persists to PostgreSQL 14 for routes and delivery receipts. Alarm Router was tested against the dependency ageing register, which records the paging provider's older application interface is still in use. Alarm Router rests on the routing table export, delivery receipts for

one quarter and two missed-page reviews so far as this review is concerned. Alarm Router was raised again with the owning engineer after the initial interview, and the position did not change.

Alarm Router sits at 51 per cent line coverage; the trend across the reviewed period was down by two points after a large refactor. Deployment of Alarm Router is cloud, regional, with a second delivery provider, which sets the blast radius of any defect in it. Reading the dependency map from Alarm Router outward gives the fleet health monitor and the on-call rotation source on the input side and the paging provider and the customer notification channel on the output side. Release cadence for Alarm Router is fortnightly, and the release records bear that out. Alarm Router is owned by the Field Reliability team, with Wanda Njoku named as the engineering owner of record. Alarm Router was one of two items where TDR-2026-05 could not obtain the underlying data within the engagement.

Fleet Allocator publishes a stable interface, though two fields used by the traffic manager, the charge scheduler and the fleet health monitor are documented only in code comments. The backlog for Fleet Allocator is 9 items deep and contains an acknowledged defect whereby assignment starvation for robots parked at chargers below the battery reserve threshold. Fleet Allocator was tested against the dependency ageing register, which records none identified in the reviewed manifests. Fleet Services releases Fleet Allocator weekly, behind a change approval, and no release in the sampled window bypassed the gate set. Hosting for Fleet Allocator is cloud, containers on a managed Kubernetes cluster, and its Go 1.22 code writes through to PostgreSQL 14 with a synchronous standby. Fleet Allocator was inspected in the repository rather than described from the ticket, which is why the account is specific.

Fleet Allocator was assessed against the objective and the defect record, and the reviewer could corroborate the availability claim but not the latency claim, which rests on a single dashboard. Fleet Allocator reports 82 per cent line coverage, measured by the same instrumentation the pipeline uses. Documents and data inspected for Fleet Allocator were the allocator service contract, two design reviews and a week of assignment audit logs. Fleet Allocator runs cloud, one regional deployment serving all sites in that region; that placement is what determines whether a fault is one robot or one region. Fleet Allocator is owned by the Fleet Services team, with Marta Kellerman named as the engineering owner of record. Fleet Allocator is reported at the evidence strength available and has not been promoted to fit the narrative.

For Battery Management Service the reviewer worked from the pack firmware release notes, cell test data for two revisions and a derating incident file. Battery Management Service was checked for shared-database coupling, and the ownership boundary held. Supportability of Battery Management Service turns on one fact: one pack revision uses a cell supplier that has since discontinued the part. Battery Management Service pairs a C on the pack controller, Python 3.11 in the cloud implementation with split: pack firmware and a cloud analytics service, and its data lives in pack non-volatile memory and a cloud time-series store. The published objective for Battery Management Service is that pack cell imbalance is reported before it reaches the derating threshold. Battery Management Service was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Reading the dependency map from Battery Management Service outward gives the pack sensors and the charge scheduler on the input side and the charge scheduler and the fleet health monitor on the output side. Robot Firmware tracks 10 open defects on Battery Management Service, including the acknowledged case where state-of-health estimates drift on packs that spend most of their life on opportunity charge. In production Battery Management Service is split edge and cloud, unchanged since the last architecture review. 54 per cent line coverage is reported for Battery Management Service, and the reviewer re-ran the measurement on a clean checkout. On paper Battery Management Service releases pack firmware twice a year, the cloud service monthly, and the observed practice did not diverge from that. Battery Management Service was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Config Distribution carries a single measurable objective, namely that a configuration change reaches every online robot inside five minutes. Config Distribution runs split edge and cloud, with a signed bundle format; that placement is what determines whether a fault is one robot or one region. Taking the evidence together, Config Distribution the component is adequate today and carries a supportability liability that is not on the funded plan. Against Config Distribution, the coverage figure is 72 per cent line coverage, which the reviewer regards that as adequate for the risk this component carries. Platform Infrastructure tracks 6 open defects on Config Distribution, including the acknowledged case where a robot that is offline at publication applies the change without re-validating preconditions. Config Distribution is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Config Distribution ships weekly; the reviewer sampled the last twelve releases. Config Distribution sits between the site provisioning service and the artefact signing service and the on-robot supervisor and the motion planner parameters, and that position is what makes its availability objective load-bearing. Config Distribution rests on the bundle format specification, distribution latency measurements and an offline apply test so far as this review is concerned. Hosting for Config Distribution is split: an on-robot agent and a cloud service, and its Rust 1.79 agent with a Go 1.22 control service code writes through to content-addressed object storage with a PostgreSQL 14 index. Config Distribution has a single named owner, Sam Oyelaran of Platform Infrastructure, which is not true of every service reviewed. Config Distribution is reported here in full because a summary of it in an earlier draft of TDR-2026-05 was misread as an assurance.

Data ownership is clear for Site Provisioning Service: it writes PostgreSQL 14 with a configuration history table and no other service writes there. Automated test coverage on Site Provisioning Service stands at 64 per cent line coverage. 5 defects remain open against Site Provisioning Service; the one the reviewer pressed on is that site templates diverged after two customers required bespoke network settings. The evidence base for Site Provisioning Service was the provisioning runbook, template diffs for four sites and two go-live records. Release cadence for Site Provisioning Service is fortnightly, and the release records bear that out. Site Provisioning Service is the entry that changed most between the first and second document requests behind TDR-2026-05.

On end-of-life exposure, Site Provisioning Service shows none identified. Hosting for Site Provisioning Service is cloud, containers on a managed Kubernetes cluster, and its Go 1.22 code writes through to PostgreSQL 14 with a configuration history table. Platform Infrastructure publishes one objective for Site Provisioning Service: a new site reaches a working configuration inside one working day. Site Provisioning Service sits cloud, global, and no alternative placement is currently supported. For Site Provisioning Service the reviewer records that the reviewer could corroborate the availability claim but not the latency claim, which rests on a single dashboard. Site Provisioning Service was cross-checked against the engineering record supplied for TDR-2026-05, and the two agree.

Safety Engineering releases Safety Controller twice a year, with full revalidation of the safety case, and no release in the sampled window bypassed the gate set. Safety Controller publishes a stable interface, though two fields used by the drive power contactors and the robot fault annunciator are documented only in code comments. Supportability of Safety Controller turns on one fact: none; the supplier's support commitment runs to 2031. 3 items sit in the Safety Controller backlog; the entry that bears on the investment case is that the fault log wraps after four thousand entries and the oldest records are discarded. The Safety Engineering team carries Safety Controller on its service inventory, and Bettina Vogel answered for it in interview. Safety Controller illustrates why TDR-2026-05 separates the severity of a finding from the confidence behind it.

The manifests place Safety Controller edge, dual-channel, with cross-checking between channels. The Safety Controller codebase is C, written to a constrained coding subset throughout, deployed onto a safety-rated controller module supplied by a third party, and its durable record is a non-volatile fault log on the controller. For Safety Controller the reviewer records that the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. 91 per cent line coverage is reported for Safety Controller, and the reviewer re-ran the measurement on a clean checkout. Safety Engineering publishes one objective for Safety Controller: a validated protective-stop trigger de-energises the drive inside 120 ms. Safety Controller was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

The Analytics Warehouse Loader codebase is Python 3.11 with SQL transformations throughout, deployed onto cloud, a scheduled batch runner, and its durable record is a columnar warehouse with daily partitions. Analytics Warehouse Loader was weighed on its coverage, its defect backlog and its dependency position, and it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Documents and data inspected for Analytics Warehouse Loader were the load schedule, partition completeness checks and two months of load audit records. Deployment of Analytics Warehouse Loader is cloud, one loader per region writing to a shared warehouse, which sets the blast radius of any defect in it. Analytics Warehouse Loader releases weekly. Analytics Warehouse Loader was excluded from an earlier draft of TDR-2026-05 and reinstated after the reviewer obtained the supporting record.

Data and Telemetry holds Analytics Warehouse Loader, and the ownership entry naming Farida Cisse matches the on-call rotation. Upstream of Analytics Warehouse Loader sit the telemetry ingest path, the order broker and the billing

usage meter; downstream, the customer reporting views and the internal metrics layer consume its output. Analytics Warehouse Loader was checked for shared-database coupling, and the ownership boundary held. Coverage for Analytics Warehouse Loader was 45 per cent line coverage at the review date, and the trend over the reviewed period was flat. Analytics Warehouse Loader commits that the daily load completes before 06:00 local time at every region, and the objective is measured continuously rather than sampled. Analytics Warehouse Loader is reported at the evidence strength available and has not been promoted to fit the narrative.

Read against its objective, Robot Shadow Store the component is adequate today and carries a supportability liability that is not on the funded plan. Robot Shadow Store releases weekly. The telemetry ingest path and the edge gateway agent feed Robot Shadow Store; a failure in Robot Shadow Store is felt by the fleet allocator, the traffic manager and the remote operations console. Robot Shadow Store runs cloud, regional, sharded by site; that placement is what determines whether a fault is one robot or one region. Robot Shadow Store is owned by the Fleet Services team, with Marta Kellerman named as the engineering owner of record. Robot Shadow Store would move from a concern to a verified defect if one further measurement were supplied.

Redis 7 as the hot path with PostgreSQL 14 as the durable record is the system of record for Robot Shadow Store, a Go 1.22 component running on cloud, containers with a sharded cache. Fleet Services publishes one objective for Robot Shadow Store: shadow state is no more than two seconds behind the robot at p95. The backlog for Robot Shadow Store is 9 items deep and contains an acknowledged defect whereby a shard rebalance briefly serves stale shadow state without marking it stale. Interface discipline around Robot Shadow Store is adequate; the schema is versioned and change is gated on a compatibility check. For Robot Shadow Store the reviewer worked from the shadow consistency note, staleness measurements and a rebalance observation. Robot Shadow Store appears twice in the working papers for TDR-2026-05 because the first answer was corrected after a follow-up request.

Schedule G - Open defect population and ageing by service (TDR-2026-05)

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Motion Planner	14	2025-10-23	2 high, 4 medium, 8 low	12	Priya Raghunathan
Costmap Fusion	21	2025-08-09	0 high, 1 medium, 20 low	1	Daniel Okafor
Fleet Allocator	9	2025-08-23	2 high, 1 medium, 6 low	6	Marta Kellerman
Traffic Manager	12	2025-09-03	0 high, 3 medium, 9 low	13	Owen Brantley
Charge Scheduler	7	2026-01-12	1 high, 3 medium, 3 low	9	Nadia Semenova
Safety Controller	3	2025-07-28	0 high, 1 medium, 2 low	12	Bettina Vogel
E-Stop Chain	2	2026-03-24	0 high, 1 medium, 1 low	4	Theo Bassett
Lidar Driver	18	2025-10-15	0 high, 3 medium, 15 low	6	Yuki Harada
Localization Engine	16	2025-08-02	1 high, 8 medium, 7 low	8	Rahul Menon
WMS Connector	23	2026-01-20	0 high, 4 medium, 19 low	5	Tomas Villareal
Order Broker	11	2026-04-18	1 high, 5 medium, 5 low	12	Grace Lindqvist
Telemetry Ingest	19	2026-01-31	1 high, 2 medium, 16 low	10	Farida Cisse
OTA Updater	8	2025-06-05	0 high, 1 medium, 7 low	11	Erik Palmgren
Identity and Authorization	6	2025-11-20	1 high, 2 medium, 3 low	11	Sonia Krall
Remote Operations Console	27	2026-03-25	1 high, 2 medium, 24 low	14	Anjali Sekhon
Simulation Harness	31	2025-05-19	2 high, 4 medium, 25 low	10	Colin Achebe
Hardware-in-the-Loop Rig	13	2025-11-09	2 high, 1 medium, 10 low	11	Nathan Oduya

4. Release cadence and ownership across the estate - part 2 of 2

Continuing the 4. Release cadence and ownership across the estate, the rows below follow directly on from part 2 minus one, with no gap in the reference sequence.

Column definitions stated with the first part of the 4. Release cadence and ownership across the estate govern the whole of it, and any qualification recorded there is carried forward here. The break is editorial.

Schedule G - Open defect population and ageing by service (TDR-2026-05) (continued, part 2 of 2)

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Fleet Health Monitor	17	2025-06-13	0 high, 5 medium, 12 low	8	Wanda Njoku
Battery Management Service	10	2026-02-01	0 high, 2 medium, 8 low	1	Peter Halvorsen
Gripper Controller	15	2025-05-28	0 high, 2 medium, 13 low	13	Ines Duarte
Map Authoring Studio	22	2025-04-02	2 high, 7 medium, 13 low	2	Lucia Ferrante
Site Provisioning Service	5	2025-11-01	1 high, 1 medium, 3 low	1	Marcus Duplessis
Event Bus	4	2025-10-13	0 high, 1 medium, 3 low	8	Erik Palmgren
Config Distribution	6	2026-03-11	1 high, 3 medium, 2 low	4	Sam Oyelaran
Robot Shadow Store	9	2025-10-18	2 high, 2 medium, 5 low	13	Marta Kellerman
Analytics Warehouse Loader	20	2025-03-05	1 high, 3 medium, 16 low	10	Farida Cisse
Billing Usage Meter	7	2025-06-01	0 high, 1 medium, 6 low	1	Nadia Semenova
Alarm Router	14	2025-12-18	0 high, 1 medium, 13 low	6	Wanda Njoku
Video Snippet Service	12	2025-06-22	1 high, 2 medium, 9 low	5	Anjali Sekhon
Speed and Separation Monitor	4	2025-06-30	1 high, 2 medium, 1 low	10	Bettina Vogel
Wheel Odometry Filter	8	2026-01-20	1 high, 3 medium, 4 low	9	Yuki Harada
Payload Scale Interface	11	2026-03-17	0 high, 4 medium, 7 low	7	Ines Duarte
Edge Gateway Agent	5	2025-04-30	1 high, 1 medium, 3 low	2	Sam Oyelaran
Release Orchestrator	16	2025-11-29	1 high, 5 medium, 10 low	5	Colin Achebe

5. Change-related incidents and restore times

Incident INC-2026-028 carries limited, being a single account corroborated only by the ticket timeline evidence, and the reviewer declined to raise the finding beyond what that supports. So far as reach is concerned, Fleet Health Monitor incident INC-2026-028 involved one site and 22 robots. Drawing the evidence in INC-2026-028 together, an operational workaround masked the defect for longer than the incident record suggests. Landed on 2026-04-10 is the position on the observability event INC-2026-028 commitment as at the review date. Reported detection applies to incident INC-2026-028, since the customer asked why one symptom persisted after the case had been closed. observability event INC-2026-028 illustrates why TDR-2026-05 separates the severity of a finding from the confidence behind it.

Fleet Health Monitor accounts for INC-2026-028, a observability event. The engineering team's root cause for observability incident INC-2026-028 is that alert grouping keyed on site and severity rather than on the underlying signal, and the reviewer found the analysis sound. The reviewer treats one contributing factor in Granite Peak Distribution incident INC-2026-028 as the more important lesson: case closure did not check for unresolved constituent alerts. Duration for observability incident INC-2026-028 splits into not a service outage; 2 days of delayed repair of impact and 31 days to close the underlying cause. The immediate event in INC-2026-028 was that two distinct faults at one site were

grouped into a single case and the second went unworked for two days, dated 2026-03-10. incident INC-2026-028 rests on a single source, and the reviewer has said so rather than implying corroboration.

On delivery, concurrency event INC-2026-039 shows this: landed on 2026-04-26. Reported detection applies to INC-2026-039, since the site supervisor reported a blocked location. After concurrency incident INC-2026-039, Grace Lindqvist took the action to serialise cancellation against assignment and add an hourly orphan sweep by 2026-05-31. Concurrency event INC-2026-039 records that on 2026-04-07, an order cancellation raced with assignment and left an orphaned task that blocked a pick face for an hour. Drawing the evidence in INC-2026-039 together, an operational workaround masked the defect for longer than the incident record suggests. Order Broker incident INC-2026-039 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

For incident INC-2026-039 the reviewer holds strong, resting on the production traces the team supplied evidence and reports the finding at that strength. INC-2026-039 was caused by a design gap: cancellation and assignment wrote to the task record without a shared lock. Incident INC-2026-039 shows a short mitigation and a long tail: 1 h 04 m against 19 days. Incident INC-2026-039 would have been shorter or smaller had it not been that the journal reconciliation runs nightly, so the orphan was invisible until the morning. Scope of impact in INC-2026-039 was one site and 1 robots, taken from the fleet records rather than from the narrative. Crestline Fulfillment incident INC-2026-039 carries a follow-up request that was satisfied after the draft of TDR-2026-05 was circulated.

INC-2025-063 belongs to the policy defect group and sits with Fleet Allocator. Fleet Allocator incident INC-2025-063 produced one commitment: to make the allocator consult charger availability before excluding a robot, owned by Marta Kellerman, dated 2025-07-31. Policy defect incident INC-2025-063 was caused by a design gap: the allocator filter excluded low-charge robots without checking whether a charge slot was actually available. Fleet Allocator incident INC-2025-063 took 6 h 05 m to stabilise and 12 days to close out. Fleet Allocator incident INC-2025-063 supports one conclusion and not another: the fix is real and the regression protection is not, so recurrence remains possible. Granite Peak Distribution incident INC-2025-063 has no single accountable owner, and that is the finding the reviewer would press hardest.

INC-2025-063 was contained to 3 sites and 28 robots. The trigger recorded for policy defect incident INC-2025-063 on 2025-05-27 was that robots parked below the battery reserve threshold were never offered work and the site ran short of movers. Policy defect event INC-2025-063 closure status is landed on 2025-06-08. Detection came through an operator noticed idle robots and raised a support ticket, which places INC-2025-063 on the reported side of the ledger. INC-2025-063 carries a second-order finding: the charge scheduler and the allocator hold separate views of charger availability. policy defect event INC-2025-063 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-05.

In taxonomy terms INC-2026-051 is a configuration drift incident in Site Provisioning Service. Incident INC-2026-051 was caused by a design gap: the site template had been forked for a customer network requirement and drifted from the baseline. The remediation committed after incident INC-2026-051 was to check every template against the baseline and re-converge the four forked templates, owned by Marcus Duplessis with a due date of 2026-06-15. The immediate event in incident INC-2026-051 was that a new site went live with a bespoke network template that omitted the outbound certificate pin, dated 2026-05-06. As to whether the configuration drift event INC-2026-051 remediation actually landed: the check landed on 2026-05-15; two templates remain forked. configuration drift event INC-2026-051 rests on a single source, and the reviewer has said so rather than implying corroboration.

Evidence strength on the configuration drift incident INC-2026-051 analysis is moderate; the narrative is consistent but the reviewer could not independently reproduce it. Mitigation of incident INC-2026-051 took 2 h 18 m and full resolution took 9 days. What configuration drift event INC-2026-051 tells the reviewer is that the failure was in the detection design rather than in the code that failed. The reviewer treats one contributing factor in INC-2026-051 as the more important lesson: template drift was not detected because the baseline check compares only the default template. Site Provisioning Service incident INC-2026-051 was detected because the identity service rejected the site's first token request. Larkspur Home incident INC-2026-051 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Environmental event INC-2026-068 traces back to a single fact: the filter has no wet-floor slip model and does not flag the condition to the operator. The two durations recorded for Harborline Systems incident INC-2026-068 are not a

service outage; 3 h of degraded operation to mitigate and open at the review date to resolve. On 2026-06-16 the record for Harborline Systems incident INC-2026-068 opens: odometry drift exceeded specification on a wet floor and localization confidence fell below the mission threshold. Drawing the evidence in environmental event INC-2026-068 together, an operational workaround masked the defect for longer than the incident record suggests. In taxonomy terms incident INC-2026-068 is a environmental incident in Wheel Odometry Filter. INC-2026-068 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

For Harborline Systems incident INC-2026-068 the follow-through was tested rather than assumed, and not landed; the detector is prototyped and unvalidated. The detection path for incident INC-2026-068 was robots stopped and requested assistance, which the site reported as unplanned downtime. The contributing factor recorded against Harborline Systems incident INC-2026-068 is that the site's cleaning schedule changed and now overlaps the peak shift. The corrective action for environmental event INC-2026-068 is to add a slip detector that flags the condition and issue a site advisory on cleaning windows, and 2026-09-15 is the date the team gave. Limited; the log retention window had closed before the reviewer asked for the raw data is the evidence grade for Harborline Systems incident INC-2026-068, and the reader should not read severity into it. environmental event INC-2026-068 was raised again with the owning engineer after the initial interview, and the position did not change.

Classifying INC-2025-131 as process bypass matters because the Map Authoring Studio backlog contains two further items of the same kind. Reported detection applies to incident INC-2025-131, since a field engineer reported unexpected keep-out zones. 2025-10-14 is the date of process bypass incident INC-2025-131, on which an unpublished map draft was exported and hand-loaded at a site and robots ran against unreviewed geometry. After Juniper Bay Apparel incident INC-2025-131, Lucia Ferrante took the action to restrict export to published revisions and shorten the publication path so the workaround is unnecessary by 2025-12-15. Assessed on its own record, Juniper Bay Apparel incident INC-2025-131 shows that management's severity rating is defensible and its remediation priority is not. Juniper Bay Apparel incident INC-2025-131 illustrates why TDR-2026-05 separates the severity of a finding from the confidence behind it.

3 h 22 m elapsed before incident INC-2025-131 was mitigated, with resolution at 44 days. Contributing to Juniper Bay Apparel incident INC-2025-131, and not repaired by the remediation, is that the site had a standing workaround for slow map publication that predated the review step. The affected population in INC-2025-131 was one site and 34 robots. Cause analysis in process bypass event INC-2025-131 concluded that the export function did not check publication state. On delivery, Juniper Bay Apparel incident INC-2025-131 shows this: the export restriction landed on 2025-11-27; the publication work is open. process bypass event INC-2025-131 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Mitigation of data retention incident INC-2025-152 took not applicable; retention only and full resolution took 37 days. Remediation for incident INC-2025-152 is to apply the lifecycle rule through the configuration baseline and delete the over-retained clips; Anjali Sekhon owns it and the committed date is 2026-01-31. The rule landed on 2026-01-10; deletion of the historical clips is open is the position on the data retention event INC-2025-152 commitment as at the review date. Reading INC-2025-152 from its first entry, on 2025-12-04 one region retained operator video clips for ninety days against a thirty-day commitment. Taken with the rest of the record, Video Snippet Service incident INC-2025-152 indicates that the failure was in the detection design rather than in the code that failed. data retention event INC-2025-152 was cross-checked against the engineering record supplied for TDR-2026-05, and the two agree.

The detection path for Video Snippet Service incident INC-2025-152 was an internal privacy review found the discrepancy. The affected population in data retention event INC-2025-152 was no production site and no robot directly, the exposure being to internal systems. Strong, resting on the production traces the team supplied is the evidence grade for incident INC-2025-152, and the reader should not read severity into it. Video Snippet Service owns incident INC-2025-152, which the reviewer classes as data retention. The post-mortem for INC-2025-152 identifies the cause as follows: the lifecycle rule was applied per storage bucket and one region's bucket was created later without it. data retention incident INC-2025-152 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Data integrity incident INC-2025-058 belongs to the data integrity group and sits with WMS Connector. INC-2025-058 took 2 h 35 m to stabilise and 21 days to close out. INC-2025-058 supports one conclusion and not another: the

team's analysis was honest about a gap it could have concealed. For data integrity incident INC-2025-058 the follow-through was tested rather than assumed, and landed on 2025-05-30; the backfill was completed on 2025-06-03. Strong, since the reviewer reproduced the failure mode in a laboratory environment is the evidence grade for Dunbar Industrial incident INC-2025-058, and the reader should not read severity into it. Dunbar Industrial incident INC-2025-058 carries a follow-up request that was satisfied after the draft of TDR-2026-05 was circulated.

Remediation for data integrity event INC-2025-058 is to key suppression on the tenant and identifier pair and backfill the affected order records; Tomas Villareal owns it and the committed date is 2025-07-15. Data integrity incident INC-2025-058 records that on 2025-05-09, two tenants reused the same order identifier namespace and duplicate suppression discarded live orders. What went wrong in Dunbar Industrial incident INC-2025-058 was that duplicate suppression keyed on a customer-supplied identifier alone rather than on tenant and identifier. Attention came to incident INC-2025-058 when the customer reported missing picks the same afternoon. Incident INC-2025-058 touched 2 site or tenant with no robot taken out of service, and Dunbar Industrial was the affected account. incident INC-2025-058 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Schedule S - Rollback and recovery drills observed or evidenced (TDR-2026-05)

Drill	Date	Component	Scope	Result	Gap identified
RD-2025-01	2025-08-29	Gripper Controller	one site	aborted	restore time above target
RD-2025-02	2025-09-15	Hardware-in-the-Loop Rig	one region	aborted	approval not recorded
RD-2025-03	2025-09-30	Telemetry Ingest	one site	succeeded	state left in the paused ring
RD-2025-04	2025-10-19	Costmap Fusion	one region	failed and was retried	approval not recorded
RD-2025-05	2025-11-01	Charge Scheduler	one ring	failed and was retried	approval not recorded
RD-2025-06	2025-11-17	Billing Usage Meter	one ring	succeeded	runbook step missing
RD-2025-07	2025-12-03	Fleet Health Monitor	one site	aborted	restore time above target
RD-2025-08	2025-12-22	Identity and Authorization	one region	aborted	state left in the paused ring
RD-2025-09	2026-01-04	E-Stop Chain	one region	succeeded with an observation	restore time above target
RD-2025-10	2026-01-23	Telemetry Ingest	one ring	succeeded	approval not recorded
RD-2025-11	2026-02-05	Safety Controller	one site	succeeded	none
RD-2025-12	2026-02-21	Alarm Router	one site	failed and was retried	state left in the paused ring
RD-2025-13	2026-03-12	Gripper Controller	one ring	failed and was retried	none
RD-2025-14	2026-03-27	Identity and Authorization	one ring	succeeded with an observation	restore time above target

5. Change-related incidents and restore times - part 2 of 2

Below, the 5. Change-related incidents and restore times continues; nothing separates this part from the last except the page on which it happens to fall. Nothing is restated from the earlier part.

Definitions given once for the 5. Change-related incidents and restore times are relied on again rather than restated, the compilation basis being identical. Pagination alone determines where it falls.

Schedule S - Rollback and recovery drills observed or evidenced (TDR-2026-05) (continued, part 2 of 2)

Drill	Date	Component	Scope	Result	Gap identified
RD-2026-01	2026-04-12	Site Provisioning Service	the full fleet in staging	failed and was retried	runbook step missing
RD-2026-02	2026-04-27	Motion Planner	one ring	failed and was retried	none
RD-2026-03	2026-05-13	Battery Management Service	one ring	succeeded with an observation	runbook step missing
RD-2026-04	2026-05-31	OTA Updater	one robot	aborted	none
RD-2026-05	2026-06-14	OTA Updater	one region	failed and was retried	none
RD-2026-06	2026-07-01	Edge Gateway Agent	one region	succeeded	none
RD-2026-07	2026-07-16	Costmap Fusion	one robot	failed and was retried	none
RD-2026-08	2026-08-01	Video Snippet Service	one ring	succeeded with an observation	state left in the paused ring
RD-2026-09	2026-08-19	Alarm Router	the full fleet in staging	succeeded	none
RD-2026-10	2026-09-03	E-Stop Chain	one region	aborted	approval not recorded

Drill	Date	Component	Scope	Result	Gap identified
RD-2026-11	2026-09-19	Robot Shadow Store	one robot	succeeded	none
RD-2026-12	2026-10-05	Site Provisioning Service	one site	succeeded	runbook step missing
RD-2026-13	2026-10-20	Costmap Fusion	the full fleet in staging	aborted	approval not recorded
RD-2026-14	2026-11-06	Charge Scheduler	one robot	succeeded	none

6. Tooling and test-infrastructure debt

Effort on Hardware-in-the-Loop Rig item TD-05 is estimated at 0.5 engineer-quarters, costed at $96,000. Hardware-in-the-Loop Rig item TD-05 carries this observation: outstanding since the April 2025 lidar timing incident. Item register item TD-05 in the debt register is to bring the second bench rig to the current sensor configuration, and it bears on Hardware-in-the-Loop Rig. TD-05 would move from a concern to a verified defect if one further measurement were supplied.

Reading Event Bus item TD-09 against the funded plan, the upgrade requires a maintenance window at every site. Event Bus item TD-09 appears in the register and is funded and already staffed. Register item TD-09 is sequenced in parallel with the release-process changes. support item TD-09 is the clearest instance in TDR-2026-05 of engineering practice being better than the documentation suggests.

Performance item TD-13 is on the register because it relocalisation exceeds the operator prompt timeout at sparse sites. Effort on TD-13 on Localization Engine is estimated at 1.5 engineer-quarters, costed at $178,000. Performance item TD-13 is the item to replace the fallback scan matcher with the current implementation, classified under performance. programme item TD-13 is reported at the evidence strength available and has not been promoted to fit the narrative.

Sizing on TD-04 on Costmap Fusion gives 2 engineer-quarters and $240,000. The reviewer's qualification on TD-04 on Costmap Fusion is that coverage moved from 51 to 58 per cent over the reviewed period. TD-04 covers the work to raise robot software line coverage above sixty per cent on the perception path. TD-04 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Charge Scheduler item TD-23 is sequenced behind a supplier release outside the company's control. The business consequence recorded against TD-23 on Charge Scheduler is that it one noisy query affects task assignment. 2026-12-31 is the target for register item TD-23, whose status is committed. programme item TD-23 was raised again with the owning engineer after the initial interview, and the position did not change.

Effort on TD-10 is estimated at 0.5 engineer-quarters, costed at $54,000. Reading register item TD-10 against the funded plan, the warehouse vendor has published a withdrawal date. Analytics Warehouse Loader carries register item TD-10, the work to migrate the deprecated warehouse transformation function. programme item TD-10 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

The plan treats Battery Management Service item TD-21 as partly funded, with the second phase unbudgeted, which the reviewer checked against the headcount model. In progress is the status of register item TD-21, targeted at 2026-09-30. Refit the battery state-of-health estimator on opportunity-charge data is entered at TD-21 on Battery Management Service as an quality item. TD-21 on Battery Management Service was compared with the equivalent position at two other components, and it is neither the best nor the worst.

The reviewer's comment on TD-19 on Speed and Separation Monitor is that the most significant open safety item in the reviewed record. What safety item TD-19 protects against is the fact that it the protective distance rests on a fixed approach speed assumption. Unfunded and dependent on capital rather than engineering effort is the funding position on TD-19 on Speed and Separation Monitor. register item TD-19 carries a follow-up request that was satisfied after the draft of TDR-2026-05 was circulated.

About safety item TD-20 it should be said that the prototype uses a network-copyleft component that must be replaced before shipping. Safety item TD-20 covers the work to validate and ship the wheel slip detector. The plan treats TD-20 on Wheel Odometry Filter as funded in the current financial year, which the reviewer checked against the headcount model. TD-20 on Wheel Odometry Filter is the clearest instance in TDR-2026-05 of a control that exists on paper and is weaker in practice.

Sizing on TD-06 gives 1 engineer-quarter and $115,000. The dependency structure places programme item TD-06 behind a supplier release outside the company's control. The quality entry at TD-06 on Simulation Harness is to make the nightly simulation suite blocking. TD-06 on Simulation Harness is the clearest instance in TDR-2026-05 of a control that exists on paper and is weaker in practice.

The business consequence recorded against TD-18 is that it investigation evidence is lost between weekly extractions. The dependency structure places programme item TD-18 behind a platform change that has its own target date. TD-18 covers the work to aggregate the safety controller fault log to the cloud continuously. TD-18 on Safety Controller rests on a single source, and the reviewer has said so rather than implying corroboration.

TD-08 is the item to replace the archived charting library in the operator console, classified under security supportability. Register item TD-08 is funded from the second quarter onward in the engineering plan the reviewer was shown. On register item TD-08 the reviewer records that no advisory affects the current version, which is why it ranks low. TD-08 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Re-converge the four forked site provisioning templates is entered at TD-11 as an configuration item. The reviewer's qualification on TD-11 is that two templates remain forked as at the review date. What TD-11 on Site Provisioning Service protects against is the fact that it template drift produced a go-live failure in May 2026. programme item TD-11 appears twice in the working papers for TDR-2026-05 because the first answer was corrected after a follow-up request.

On funding, quality item TD-22 is funded only if the team's headcount plan is met. If TD-22 on Payload Scale Interface is not done, the exposure is that it calibration is lost on a compute module swap. TD-22 on Payload Scale Interface is not straightforward: a small item that keeps returning as a field support cost. quality item TD-22 appears twice in the working papers for TDR-2026-05 because the first answer was corrected after a follow-up request.

7. Findings in the build and release path

After the fix, SEC-2025-08 against Event Bus carries residual risk described as low; the interface is restricted to the operations subnet. Remediated is the status of SEC-2025-08, dated 2025-10-31. 2025-09-23 is the discovery date of assessment finding SEC-2025-08, which finds that event bus administrative interface reachable from the application subnet. On assessment finding SEC-2025-08 the tester's evidence was that reachability proven by the tester from a deployed container, and the reviewer grades it strong. finding SEC-2025-08 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Simulation Harness finding SEC-2026-06 matters because a compromised scenario could read production artefact storage. On 2026-04-17 the tester recorded assessment finding SEC-2026-06: simulation harness cluster used a long-lived cloud role with broad read access. For assessment finding SEC-2026-06 the reviewer notes that the severity is right and the ownership is unclear, since two teams each believe the other holds it. Evidence behind finding SEC-2026-06 is that role policy read from the cloud configuration export, which the reviewer treats as strong. Medium-severity finding SEC-2026-06 was reviewed against the position twelve months earlier, and nothing measurable has changed.

SEC-2026-09 is supported by the fact that service restart induced by the tester's query; evidence strength is strong. Remediated is the status of finding SEC-2026-09, dated 2026-05-08. Fleet Health Monitor finding SEC-2026-09 leaves residual risk that the reviewer assesses as low. On 2026-04-18 the tester recorded assessment finding SEC-2026-09: fleet health monitor query interface allowed unbounded result sets. Fleet Health Monitor finding SEC-2026-09 carries a follow-up request that was satisfied after the draft of TDR-2026-05 was circulated.

High-severity finding SEC-2025-01 was opened on 2025-09-16 and states that service tokens are issued with a twelve-hour lifetime. Assessment finding SEC-2025-01 produces this judgement: the underlying pattern appears in two other components that the tester did not reach. SEC-2025-01 against Identity and Authorization reduces to a residual risk of low once lifetimes are cut to one hour with refresh. Remediated is the status of Identity and Authorization finding SEC-2025-01, dated 2025-11-14. finding SEC-2025-01 carries a follow-up request that was satisfied after the draft of TDR-2026-05 was circulated.

SEC-2026-10 against Billing Usage Meter stands remediated; 2026-06-05 is what the record gives for closure. SEC-2026-10 against Billing Usage Meter leaves residual risk that the reviewer assesses as low; object lock is applied

and verified nightly. Billing Usage Meter finding SEC-2026-10 rests on the observation that bucket configuration read from the cloud export; the reviewer holds that to be strong. The tester scored Low-severity finding SEC-2026-10 at 4.0 and labelled it Low. Low-severity finding SEC-2026-10 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Accepted is the status of SEC-2025-12, dated not planned. The reason SEC-2025-12 against Analytics Warehouse Loader was not downgraded is that a loader defect or compromise could alter source telemetry. SEC-2025-12 was graded Low, band 4.2, by Redwater Security engagement RW-2025-3. Reading SEC-2025-12 against Analytics Warehouse Loader, the defect is that analytics warehouse service account held write access to raw telemetry; the discovery date is 2025-09-25. finding SEC-2025-12 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

What SEC-2025-04 enables is this: an authenticated author can distribute unreviewed geometry. On SEC-2025-04 the conclusion is that the retest evidence is a screenshot rather than a re-run of the original test, which the reviewer notes. As at the review date assessment finding SEC-2025-04 was remediated and the closure entry reads 2025-11-27. Reading Medium-severity finding SEC-2025-04, the defect is that map authoring export bypasses the publication review; the discovery date is 2025-09-18. Map Authoring Studio finding SEC-2025-04 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

SEC-2025-02 against Release Orchestrator is remediated, with a closure date of 2025-09-19. On SEC-2025-02 the conclusion is that the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. On High-severity finding SEC-2025-02 the tester's evidence was that the tester extracted a working token from a retained log, and the reviewer grades it moderate. High-severity finding SEC-2025-02 translates into the following exposure: any engineer with build log access could replay a service credential. assessment finding SEC-2025-02 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Read alongside the remediation record, Motion Planner finding SEC-2026-17 shows that the severity is right and the ownership is unclear, since two teams each believe the other holds it. 2026-04-22 is the discovery date of Motion Planner finding SEC-2026-17, which finds that motion planner parameter override accepted from an unauthenticated local port. What Motion Planner finding SEC-2026-17 enables is this: a local compromise could alter planner limits without detection. High-severity finding SEC-2026-17 arrives as a High finding with a 7.9 score, and the reviewer did not adjust either. finding SEC-2026-17 was excluded from an earlier draft of TDR-2026-05 and reinstated after the reviewer obtained the supporting record.

For SEC-2025-03 the record shows that demonstrated in the tester's laboratory network, not on a customer site, which is moderate evidence and is graded separately from severity. SEC-2025-03 against Remote Operations Console leaves residual risk that the reviewer assesses as low; pinning is applied at the gateway and the agent. Finding SEC-2025-03 translates into the following exposure: an on-path attacker on a customer network could observe operator commands. Assessment finding SEC-2025-03 is assessed as follows: the underlying pattern appears in two other components that the tester did not reach. SEC-2025-03 illustrates why TDR-2026-05 separates the severity of a finding from the confidence behind it.

8. Performance gates and their calibration

The environment behind load run LT-2025-15 carries production-sized data but a single availability zone, and the reviewer weighted the result accordingly. Weighed against the projected fleet, run LT-2025-15 indicates that the error rate rather than the latency is the term that would first breach a customer commitment. Configuration for LT-2025-15 was 480 robots against 9,400 orders an hour in the four sites, shared region case. 480-robot run LT-2025-15 carries a follow-up request that was satisfied after the draft of TDR-2026-05 was circulated.

The constraint surfaced by LT-2025-16 is shadow store shard hotspotting on the two largest sites. Six sites, shared region run LT-2025-16 is best read against the neighbouring run in the same series, from which it differs in the presence of a failure injection. Before reading LT-2025-16, note that the environment carries production data volumes and a stubbed paging provider. six sites, shared region run LT-2025-16 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Order burst, ten times nominal for two minutes run LT-2025-17 returned 620 ms median, 2,350 ms at the 95th percentile and 5,900 ms at the 99th, erroring at 1.24%. Run LT-2025-17 exercised the order burst, ten times nominal for two minutes case with 240 robots and 12,000 orders an hour, on 2025-10-21. On order burst, ten times nominal for two minutes run LT-2025-17 the reviewer notes that the pass verdict depends on a target that was last recalibrated in 2024. LT-2025-17 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Before reading LT-2025-18, note that the environment carries a replayed order stream taken from the two largest sites. In telemetry burst with control traffic on the shared topic run LT-2025-18 the bottleneck was reproduced the November 2025 control delay before the topic split. The size of the concurrent operator session pool is what separates LT-2025-18 from the adjacent run. 480-robot run LT-2025-18 is the clearest instance in TDR-2026-05 of engineering practice being better than the documentation suggests.

Fail against the error rate target is what the test record gives for region failover during peak run LT-2025-19. Configuration for load run LT-2025-19 was 480 robots against 9,400 orders an hour in the region failover during peak case. Scenario LT-2025-19 is not read as a headline number: the run was executed against a synthetic warehouse system rather than a customer endpoint, which flatters the tail. 480-robot run LT-2025-19 was excluded from an earlier draft of TDR-2026-05 and reinstated after the reviewer obtained the supporting record.

The limiting term in scenario LT-2025-20 was recovery inside 40 s but with an assignment gap. The point of 240-robot run LT-2025-20 within the series is the presence of a failure injection. Measured latency in database failover, allocator primary run LT-2025-20 was 340 ms at the median, 1,410 ms at p95 and 4,200 ms at p99. LT-2025-20 was excluded from an earlier draft of TDR-2026-05 and reinstated after the reviewer obtained the supporting record.

Configuration of the environment for scenario LT-2026-01 included production-sized data but a single availability zone. The reviewer's reading of 720-robot run LT-2026-01 is that one run does not establish a trend, and this scenario was executed once. Scenario LT-2026-01 shows where the system runs out of headroom: the control topic split removed the telemetry coupling. six sites after the topic split run LT-2026-01 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Weighed against the projected fleet, eight sites, shared region run LT-2026-02 indicates that the failure is a capacity finding rather than a defect, and the remediation is a scheduled engineering programme. 960 robots and 18,200 orders an hour define scenario LT-2026-02, run on 2026-02-12. Measured latency in run LT-2026-02 was 455 ms at the median, 1,340 ms at p95 and 2,880 ms at p99. LT-2026-02 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Load run LT-2026-03 sits against the 1,500 ms p95 acknowledgement objective for the warehouse connector; whether that target is the right one is a separate question. Scenario LT-2026-03 came out pass. The eight sites with the allocator read replicas enabled run at load run LT-2026-03 was executed on 2026-02-13 at 960 robots and 18,200 orders an hour. run LT-2026-03 rests on a single source, and the reviewer has said so rather than implying corroboration.

Against the stated targets, 1200-robot run LT-2026-04 is fail against the 1,200 ms p95 target. Run LT-2026-04 is the ten sites, shared region run, sized at 1200 robots and 22,600 orders an hour. LT-2026-04 identified the binding constraint: the allocator write path needs partitioning to go further. scenario LT-2026-04 illustrates why TDR-2026-05 separates the severity of a finding from the confidence behind it.

1,240 ms at p95 is the figure that matters in load run LT-2026-05, against 480 ms median and 2,760 ms at the tail. The target scenario LT-2026-05 was measured against is the internal objective that no telemetry is lost for an outage under four hours. Load run LT-2026-05 was designed to change one variable, namely the order rate rather than the fleet size. LT-2026-05 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Shard rebalance under peak load run LT-2026-06 used the standard performance environment, which is characterised by the current production topology with one region held out. Shard rebalance under peak load is the scenario at LT-2026-06: 720 robots, 13,800 orders an hour, dated 2026-02-19. Scenario LT-2026-06 is recorded as pass with observation. load run LT-2026-06 appears twice in the working papers for TDR-2026-05 because the first answer was corrected after a follow-up request.

On 2026-02-20 the team ran load run LT-2026-07, the identity service token storm case, at 960 robots. In load run LT-2026-07 the median was 180 ms and the tail reached 1,010 ms; errors ran at 0.03%. The point of 960-robot run LT-2026-07 within the series is one platform change between otherwise identical runs. identity service token storm run LT-2026-07 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Load run LT-2026-08 is not read as a headline number: the test environment carried a smaller dataset than production, so the database term is understated here. Outcome of LT-2026-08 was pass, and the reviewer read the raw result rather than the summary. Measurement of scenario LT-2026-08 is against the internal objective that no telemetry is lost for an outage under four hours rather than against a customer commitment. scenario LT-2026-08 was raised again with the owning engineer after the initial interview, and the position did not change.

In the series run LT-2026-09 is the run that changes one platform change between otherwise identical runs. Load run LT-2026-09 is recorded as fail. 480 robots and 9,400 orders an hour define run LT-2026-09, run on 2026-02-25. 480-robot run LT-2026-09 rests on a single source, and the reviewer has said so rather than implying corroboration.

9. How a release reaches the sites

Capacity headroom at Lockbourne, Ohio is 68%, and the warehouse system is vendor A. Site ST-17 is Lockbourne, Ohio: Granite Peak Distribution, 19 robots, live from 2026-04. Lockbourne, Ohio was checked against the corresponding entry in the company's own register, and the register was current.

What distinguishes Tracy, California is that it is the site whose cleaning schedule now overlaps the peak shift. Capacity headroom at Tracy, California is 36%, and the warehouse system is vendor B. Tracy, California is treated as material to the investment case rather than as a routine engineering item.

What distinguishes Newnan, Georgia is that it is narrow-aisle geometry, and the source of the August 2025 mutual-yield stall. Remaining capacity at Newnan, Georgia is 33%, measured against the site's peak order profile. Newnan, Georgia sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

71% of headroom remains at Mesquite, Texas, whose connector targets vendor C. 2026-06 is the go-live month at Mesquite, Texas, where Evergreen Warehousing now runs 16 robots. Mesquite, Texas is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Wilmington, Ohio was included in the sample because it is the largest deployment and the closest to the allocator scaling limit. Against its peak profile, Wilmington, Ohio retains 14% headroom on a vendor B integration. Wilmington, Ohio was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Fathom Mobility operates 47 Helioscope robots at Shepherdsville, Kentucky, commissioned in 2025-03. Vendor C is the warehouse system at Shepherdsville, Kentucky, where headroom is 41%. Shepherdsville, Kentucky was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Ontario, California reports 26% headroom against its integration with vendor C. Ontario, California is the site whose shift pattern change caused the April 2025 charge incident, which is why the reviewer sampled it. Ontario, California was cross-checked against the engineering record supplied for TDR-2026-05, and the two agree.

At Meridian, Idaho the fleet is 96 robots, and the site has been live since 2022-09. Against its peak profile, Meridian, Idaho retains 22% headroom on a vendor A integration. Meridian, Idaho was cross-checked against the engineering record supplied for TDR-2026-05, and the two agree.

Internal engineering site operates 34 Helioscope robots at Fremont, California, commissioned in 2021-06. The reason Fremont, California appears here is that it is the reference site, and the only one with a laboratory alongside. Fremont, California is the only entry in its group where the reviewer and the engineering team reached different conclusions.

The Grand Prairie, Texas deployment serves Asterion Logistics with 78 robots, live since 2023-04. Headroom at Grand Prairie, Texas stands at 31%; vendor A is the integrated warehouse system. Grand Prairie, Texas was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Bethlehem, Pennsylvania is sparse survey features, which is why relocalisation is slow here, which is why the reviewer sampled it. 2025-11 is the go-live month at Bethlehem, Pennsylvania, where Kestrel Medical Supply now runs

37 robots. Bethlehem, Pennsylvania would move from a concern to a verified defect if one further measurement were supplied.

Capacity headroom at Edwardsville, Illinois is 63%, and the warehouse system is vendor B. For Edwardsville, Illinois the material fact is that it is the newest Crestline site, still ramping. Edwardsville, Illinois was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Capacity headroom at Lehigh Valley, Pennsylvania is 38%, and the warehouse system is vendor B. 64 robots are deployed at Lehigh Valley, Pennsylvania, which went live in 2024-02. Lehigh Valley, Pennsylvania was one of two items where TDR-2026-05 could not obtain the underlying data within the engagement.

What distinguishes Chino, California is that it is the go-live delayed by the forked provisioning template. Chino, California came online in 2026-01 and now carries 33 robots for Larkspur Home. Chino, California illustrates why TDR-2026-05 separates the severity of a finding from the confidence behind it.

Kenosha, Wisconsin was included in the sample because it is the second opportunity-charging site. 2024-11 is the go-live month at Kenosha, Wisconsin, where Dunbar Industrial now runs 41 robots. Kenosha, Wisconsin was checked against the corresponding entry in the company's own register, and the register was current.

10. Process findings management does not accept

The report at TDR-2026-05 records 2 items the engineering team declined to remediate, each with the reason given and the reviewer's view of it. The reviewer asked in each case whether a declined item had a compensating control, and this report reports the answer even where it is unsatisfactory. The sample behind this report is 9 services and 7 sites; the selection was the reviewer's, not the company's. K. Vandermeer chose the sample for this report after reading the dependency map, so the selection is not proportionate to headcount or to code volume. K. Vandermeer interviewed 12 people for TDR-2026-05; where an account was uncorroborated it is marked as such.

K. Vandermeer has dated each open item in report TDR-2026-05 so that a reader can see which have been open longest rather than only how many there are. The report at TDR-2026-05 closes with 8 open items, listed with owner, evidence and the reviewer's view of whether the committed date is realistic. Evidence gathering for TDR-2026-05 was iterative: an initial document request, 12 interviews, then a second request that closed most of the gaps. The approach K. Vandermeer took in report TDR-2026-05 was to test the company's own assertions rather than to restate them. This report deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.

This report reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. K. Vandermeer treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and report TDR-2026-05 keeps the two apart. This report is one of a series of technical reports; this one takes the engineering delivery process, pipeline metrics and change-management controls and reports against the evidence available by 2026-03-03. 2026-03-03 is the evidence cut-off for this report, whose subject is the engineering delivery process, pipeline metrics and change-management controls. The engineering delivery process, pipeline metrics and change-management controls produced more evidence than the present report reproduces, and the working papers hold the remainder.

K. Vandermeer notes in TDR-2026-05 that the company's own reporting is adequate for three of the four recommended indicators. Monitoring recommended by report TDR-2026-05 relies on metrics the company already emits, so no new instrumentation is required to start. The working method behind the report at TDR-2026-05 was to start from the incident record and work back to the design, rather than the other way round. In report TDR-2026-05 the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. K. Vandermeer has kept the working papers behind TDR-2026-05 in a form that a second reviewer could re-perform.

11. Conclusions on delivery capability

K. Vandermeer would summarise the report at TDR-2026-05 in one sentence: the technical estate would support the business plan through the next twelve months and not through the fleet size the plan assumes for 2027 without the identified programme. K. Vandermeer's conclusion in the present report is that the technical estate would support the business plan through the next twelve months and not through the fleet size the plan assumes for 2027 without the

identified programme. 12 interviews inform the report at TDR-2026-05. Where an interview is the sole source for a claim, the claim is graded limited. Interviews for TDR-2026-05 were conducted with the owning engineer present rather than with a manager speaking on their behalf. K. Vandermeer has resisted rounding the findings towards a comfortable conclusion in either direction.

Deal implications in TDR-2026-05 are modest in aggregate and specific in detail: 3 items for the closing checklist, 4 for the plan. TDR-2026-05 suggests one governance change rather than a price adjustment: a quarterly technical review with the lead investor's observer present. Nothing in TDR-2026-05 rests on a test the reviewer designed; the tests were the company's, and the reviewer read the results and the method. This report is limited by log retention: the telemetry window is ninety days, so events before 2025-11-15 could not be examined directly. K. Vandermeer has separated what the company has built from what the company can demonstrate about what it has built.

The monitoring pack TDR-2026-05 proposes would have surfaced two of the incidents examined here before a customer noticed. The report at TDR-2026-05 recommends that the remediation commitment register be reported to the board quarterly with evidence attached, not just status. In the present report the reviewer records the basis of each conclusion inline, so a reader can disagree with the inference without re-doing the work. The present report distinguishes three positions: verified defect, credible concern, and matter requiring further work. The third category is not empty. K. Vandermeer has kept the company's terminology even where a different word would have read better, so that a reader can search the source documents.